

Cybersecurity Incident Report

Network Traffic Analysis

Part 1: Summary of the problem found in the tcpdump log

The tcpdump log shows the browser using the DNS protocol over UDP to contact the DNS server at 203.0.113.2 and retrieve the IP address for yummyrecipesforme.com. Instead of a valid response, the server replied over ICMP with the error "udp port 53 unreachable" for every attempt. Since port 53 is the standard port for DNS, this points to a problem with the DNS service, and it is highly likely that the DNS server is not responding on that port.

Part 2: Analysis of the data and likely cause of the incident

The incident occurred today at 13:24, when the first failed DNS query was captured, followed by repeated failures at 13:26 and 13:28 as the browser retried. Customers of the client company first reported the issue after receiving a "destination port unreachable" message when attempting to visit yummyrecipesforme.com. The cybersecurity team confirmed the problem by loading the site directly, then ran packet sniffing tests using tcpdump. In each log event, the outgoing UDP query from the client at 192.51.100.15 appears first, followed by the ICMP error response from the DNS server, which includes the flags on the UDP message (the plus sign after query identification number 35084) and the "A?" symbol indicating a request for the domain's A record.

A key finding is that the DNS server is still reachable at the network layer, since it is generating the ICMP error responses, yet nothing is listening on port 53 to answer the queries, so DNS resolution never completes and the browser cannot reach the website. The next step is to determine whether the DNS server is down, whether traffic to port 53 is blocked by the firewall, or whether a misconfiguration stopped the DNS service. The most likely cause is that the DNS server is down due to a successful Denial of Service attack or a misconfiguration.